WORLDWEAVE

GAME DESIGN DOCUMENT

Version 0.1 — Implementation Foundation

An elemental ecosystem sandbox where players learn the laws of a living world, experiment with natural forces, and leave lasting environmental consequences.

> **Document status: Draft for prototype planning. The Worldweave Version 1 concept remains the creative baseline; this GDD translates it into testable design and technical decisions.**

Document field	Definition
Working title	Worldweave (name availability not verified)
Version	GDD 0.1
Genre	Elemental ecosystem sandbox / survival / exploration / construction
Platform	Windows PC target
Engine	Unreal Engine 5.x; exact minor version to be locked at project setup
Initial mode	Single-player prototype; multiplayer is out of scope for the first playable
Core design pillar	Actions alter systems; systems create consequences; consequences create new options
Audience	Players who enjoy discovery, simulation, exploration, and creative problem-solving

Numeric values marked as targets are hypotheses to test, not final balance commitments.

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> **Sections 1–10 define player-facing design. Sections 11–15 translate it into an implementation plan. Appendix A defines the smallest useful content set for the first prototype.**

1. Executive summary and product definition

1.1 High concept

Worldweave is a third-person elemental ecosystem sandbox. The player enters a damaged, unfamiliar world and learns to influence it through four fundamental forces: fire, water, earth, and air. These forces are not merely combat spells. They alter temperature, moisture, soil, airflow, erosion, plant growth, creature routines, material states, and the feasibility of building or travelling through a region.

1.2 Player promise

> **Learn how the world works. Use that knowledge to change it. Live with the consequences—and discover opportunities that did not exist before.**

1.3 Product definition

Attribute	Definition
Genre	Exploration-led survival sandbox with systemic environmental interactions and modular construction.
Camera	Third-person for the initial prototype; first-person option remains open.
Core fantasy	Become a capable interpreter and shaper of a living landscape, not simply its strongest fighter.
Session structure	Short experiments produce immediate feedback; longer expeditions and regional projects produce persistent changes.
Primary reward	New knowledge, routes, materials, ecological opportunities, and visible world transformation.
Failure tone	Consequences should teach. Early experimentation is recoverable; high-risk actions can cost time, supplies, or access.
Initial scope	One valley, one player, four basic elements, limited interactions, a few organisms, simple shelter, and save/load.
Excluded from first prototype	Online multiplayer, procedural continent generation, complex NPC societies, large-scale terrain deformation, full fluid dynamics, AI-dependent gameplay.

1.4 Differentiation

- Elemental interactions modify the environment and resource flows, not only enemy health.
- The world retains selected consequences through a lightweight ecological-memory model.
- Creatures and plants react to understandable environmental conditions.
- Player knowledge becomes practical capability: a discovered rule can open a route, improve a build site, or solve a survival problem.
- A field journal records observations and confirmed rules without revealing every answer.

1.5 Design boundary

The game does not promise physically perfect weather, fluid simulation, terrain deformation, or creature intelligence. It delivers consistent, legible approximations that create convincing cause and effect. Players should be able to reason about outcomes, even when the result is surprising.

2. Design pillars and player experience

Pillar	Required experience	Anti-goal
Causal world	Actions produce readable, explainable effects.	Random effects that cannot be understood or repeated.
Experimentation	Players test ideas, observe results, revise hypotheses.	One correct recipe for every problem; punishing early experimentation.
Ecological memory	A region responds to accumulated changes over time.	Cosmetic-only persistence.
Knowledge is power	Discovery creates new options beyond stat increases.	Progression dominated by repetitive XP grinding.
Useful construction	Structures manage resources, shelter, access, and environmental processes.	Decorative building disconnected from gameplay.
Scope-conscious simulation	Systems interact while remaining performant.	Simulating every leaf, droplet, or creature continuously.

2.1 Intended emotional rhythm

Moment	Feeling	Design support
Notice	Curiosity	Distinct cues and unusual local conditions.
Hypothesise	Agency	Journal notes and visible clues.
Experiment	Anticipation	Low-cost early tests and clear targeting.
Understand	Satisfaction	Cause/effect confirmation and journal record.
Apply	Competence	Same rule solves a different practical problem.
Return later	Attachment	Persistent regional state and visible recovery/change.

2.2 Core loop

> **Observe → Form a hypothesis → Apply an element or combination → Read the result → Record knowledge → Use the new option → Revisit a changed region.**

2.3 Secondary loops

- Survival: scout → collect/cultivate → manage exposure and shelter → expand safe operating range.
- Construction: identify a need → survey site → gather materials → place structure → connect it to a useful process → verify.
- Creature: observe signs → infer needs/territory → change conditions or behaviour → avoid, redirect, befriend, or defend.
- Regional: diagnose a local ecological problem → alter variables → wait for bounded updates → verify stabilisation.

2.4 Agency rule

Where practical, a challenge should admit at least two valid approaches—for example, redirect water with earthworks, reduce heat exposure with shade and moisture, or find another route. A single intended solution is acceptable for a tutorial, not as the default sandbox structure.

3. World lore and narrative framework

3.1 Setting premise

The world's natural systems once operated in a stable pattern called the Weave: a network of relationships among heat, water, wind, mineral ground, living organisms, and seasonal cycles. The Weave is not a literal glowing grid; it is a term surviving cultures use to describe interdependence.

A past event known provisionally as the Sundering disrupted those relationships. Some valleys dried, some basins flooded, some forests stopped renewing, and certain creatures changed migration and territorial behaviour. The world is not dead; it is unstable and unevenly recovering. The player arrives after the initial catastrophe, when local systems can still be understood and influenced.

3.2 Narrative stance

- Tell history through environmental evidence, ruins, tools, altered vegetation, migration traces, and conflicting field notes.
- Avoid a long opening cinematic that explains every mystery; let players learn through interaction.
- Historical sites demonstrate how prior societies managed elemental processes and how misuse created consequences.
- Avoid a single mandatory moral interpretation: the player may stabilise, redirect, exploit, or reshape a local ecosystem.

3.3 Player role

The player is provisionally called a Wayfinder: an explorer who recognises patterns in the damaged world. They are not immediately a chosen saviour and do not begin with mastery of all elements. Capability is earned through observation, experimentation, and practical work.

3.4 Four elements in lore

Element	Cultural interpretation	Natural domain
Fire	Transformation; released or concentrated energy.	Heat, ignition, drying, cooking, material transformation.
Water	Continuity; movement and storage of life-supporting resources.	Moisture, runoff, pooling, cooling, transport, growth support.
Earth	Structure; the framework that holds or redirects systems.	Soil, mineral material, stability, barriers, erosion, foundations.
Air	Exchange; movement connecting separated places.	Wind, ventilation, dispersal, drying, lift, sound and airborne particles.

3.5 Narrative delivery

Channel	Purpose	Priority
Environmental storytelling	Primary source of history and regional identity.	High
Field journal	Observations, discoveries, competing hypotheses.	High
Ruins and artifacts	Prior engineering and consequences of misuse.	Medium
NPC dialogue	Partial, biased interpretations and optional tasks.	Later
Cinematics	Small number of high-value transitions.	Low for prototype

3.6 Lore guardrails

Do not fully explain the Sundering in the first region. Keep multiple plausible causes available until the wider narrative is designed. "Worldweave," "Sundering," and "Wayfinder" are working terms.

4. World structure and environmental simulation

4.1 World layout

The full-game world is divided into distinct regions or zones rather than requiring one seamless, enormous map. Each region has an ecological identity, a small set of dominant problems, unique materials, and a reason to revisit it. Zone transitions support loading, pacing, performance control, and future expansion.

Layer	Purpose	Prototype treatment
World	Overall setting, history, long-term progression.	One region only.
Region / zone	Coherent ecological and gameplay unit.	One small valley with visible boundaries.
Local cell	Bounded area for aggregated environmental variables.	Grid/spatial bins; test cell size.
Actor / object	Player, plant, creature, structure, elemental target.	Only relevant actors update frequently.

4.2 Regional state variables

Variable	Range / representation	Influences	Gameplay use
Temperature	0–100 abstract units	Fire, shade, weather, time.	Ignition, comfort, drying, habitat.
Moisture	0–100	Rain, water placement, evaporation, soil.	Growth, mud, water availability, fire resistance.
Soil stability	0–100	Ground type, flow, earthworks, vegetation.	Erosion risk, foundations, path quality.
Vegetation health	0–100 per patch/type	Moisture, temperature, soil, disturbance.	Harvesting, regrowth, habitat.
Airflow	Direction + intensity tier	Weather, terrain channels, barriers.	Smoke, mist, drying, lift.
Ecological pressure	0–100 per patch/region	Harvesting, fire, creature activity, recovery.	Regrowth rate, scarcity, migration pressure.

4.3 Simulation cadence

- Immediate: elemental hits, ignition, splashes, local state changes, and feedback update immediately.
- Short-range: nearby cells and active organisms update on a timed cadence; initial target 1–5 updates/second for relevant actors, subject to profiling.
- Regional: distant vegetation, creature populations, and recovery update in batches.
- Dormant: unloaded regions store summary state and use bounded catch-up rules on revisit or load.

4.4 Ecological memory

Ecological memory stores selected current-state changes that influence future outcomes. It is not a complete history of every event. Save current state and a few useful trend indicators rather than an unlimited event log.

- Repeated overharvesting lowers local vegetation health and increases recovery time.
- Water diversion changes moisture distribution; one route may green while another dries.
- A burned patch follows a recovery stage rather than instantly returning to baseline.
- Creature territory suitability changes when water or food patches change.

4.5 State transitions

Use explicit thresholds and discrete stages: Healthy → Stressed → Dormant/Damaged → Recovering → Healthy. Do not create a unique visual state for every numeric value. Discrete tiers are cheaper, easier to debug, and easier for players to read.

4.6 Weather and physical limits

Use authored presets—Calm, Windy, Rainfall, Dry Heat—not a full climate simulation. Presets make bounded changes to airflow, moisture, and temperature. Water movement is represented by simplified flow links/channels and cell-to-cell transfer, not computational fluid dynamics. Large-scale real-time landscape sculpting is out of scope; earth actions modify authored terrain or place discrete earthworks.

5. Elemental system: rules and interactions

5.1 System principles

- Each element has a primary action family and a small set of predictable effects.
- Effects act on tagged targets through shared interfaces, not duplicated object-specific logic.
- Interactions use explicit rules and priorities; the same conditions should produce the same gameplay result.
- VFX communicate the result but do not determine gameplay state.
- Rules are discovered through observation, the journal, and repeatable experiments.

Element	Core verbs	Targets	Risks / trade-offs
Fire	Ignite, heat, dry, cook, transform.	Combustible material, heat-sensitive resources, creatures.	Spread, smoke, resource loss, heat exposure.
Water	Wet, cool, pool, redirect, nourish.	Ground, plants, heat, channels, materials.	Mud, erosion, flooding, reduced fire effectiveness.
Earth	Raise, block, stabilise, expose, shape.	Soil, loose deposits, barriers, foundations.	Blocked routes, unstable cuts, cost, redirected water.
Air	Push, ventilate, disperse, dry, lift.	Smoke, mist, particles, light objects, thermal zones.	Fire/spore spread, dust, strong-wind loss of control.

5.2 Element application model

Player selects an unlocked element and applies it through a short readable action. The input presentation (aimed projection, placed field, or tool-like ability) remains open; the underlying API should support validation, costs, resolution, feedback, and discovery regardless of input method.

Step	Rule
Select	Show selected element and availability.
Target	Trace/query finds eligible target within range and line-of-sight rules.
Validate	Check target tags, state, cost, cooldown, and resistance.
Resolve	Apply deterministic state changes and emit an interaction event.
Feedback	Animation, VFX, audio, UI, and object state show what changed.
Record	Novel or useful results can add journal observations or confirmed rules.

5.3 Initial interaction rules

Interaction	Preconditions	Prototype outcome	Player use
Water + Fire → Steam	Water meets sufficiently hot surface/heat zone.	Brief steam volume, local cooling, temporary visibility reduction.	Cool surfaces, obscure a short route, reveal airflow.
Water + Earth → Living Soil	Moisture reaches compatible soil with organic matter.	Growth suitability rises within a cap.	Recover depleted patch, cultivate resource.
Fire + Air → Thermal Lift	Controlled heat source; airflow not too strong.	Bounded updraft force or heat transport.	Reach ledge with suitable equipment, ventilate space.
Air + Earth → Dust/Erosion	Loose dry material exposed to sufficient airflow.	Dust cloud and/or gradual stability loss.	Reveal wind direction, uncover feature, create hazard.
Fire + Earth → Altered Mineral	Heat reaches compatible deposit/material.	Material changes state or yields specialised material.	Unlock crafting or component upgrade.
Water + Air → Mist	Moisture and airflow/weather permit dispersal.	Mist patch or bounded moisture transfer.	Conceal movement, cool area, support certain plants.

5.4 Special states

State	Meaning	Prototype status
Resonance	Rare stable overlap of multiple elemental conditions.	Defer until base rules work.
Life	Compatible conditions for biological growth/recovery.	Initially derived from moisture/soil/temperature thresholds.
Phase	Authored material state changes, such as heat-softening or liquid/vapour-like effects.	Only for bounded materials; no universal phase simulator.

5.5 Conflict handling

- Safety-critical changes resolve before cosmetic effects.
- Each target has one authoritative state per state family; overlap follows declared priority.
- Effects stack only when explicitly allowed; otherwise refresh duration or apply stronger value.
- Clamp values and use cooldowns to prevent runaway feedback loops.
- Every rule logs target, rule ID, preconditions, before/after state, and rejection reason in development builds.

5.6 Readability requirement

Every meaningful result must be inferable from visual cues, audio, journal, or inspection. Invisible numerical changes without a perceptible result are not acceptable except for deliberately signposted mysteries.

6. Core gameplay loop and survival systems

6.1 Exploration

Exploration is driven by environmental questions, landmarks, resource needs, blocked routes, unusual organism behaviour, and ruins demonstrating elemental principles. Include one route that becomes easier after learning a rule, proving that knowledge has a practical payoff.

6.2 Survival needs

Need	Prototype rule	Guardrail
Health	Reduced by hazards, hostile encounters, severe exposure; restored by rest or treatment.	Signal danger before severe penalties.
Stamina	Limits sprinting, climbing, heavy actions; recovers with rest.	Keep traversal responsive.
Temperature exposure	Local heat/cold and shelter change exposure over time.	Use comfortable / stressed / critical states.
Hydration	Optional; if included, generous depletion and clear sources.	Do not add unless it creates decisions.
Hunger	Deferred until core elemental loop is validated.	Food should support the loop, not become busywork.

Implement health, stamina, and temperature exposure first. Add hydration only if it creates meaningful decisions in the valley test. Defer hunger until playtests show it strengthens rather than distracts from the elemental loop.

6.3 Hazards

- Fire/heat zones: visible cues, escalating exposure, recoverable retreat.
- Unstable ground: material cues and warning before a fall or route failure.
- Water hazards: authored volumes and rules, not full fluid simulation.
- Creatures: readable territory and warning behaviour; avoid unavoidable ambushes early.
- Weather: tactical variation without randomly erasing progress.

6.4 Death and recovery

Prototype default: respawn at a known safe point with health restored and journal knowledge retained. Whether carried resources are dropped or partially lost remains open. Early prototype tests learning and ecosystem feedback, not punitive loss.

6.5 Objective layers

- Immediate: understand one elemental interaction.
- Short-term: reach a landmark, establish shelter, or secure a resource.
- Regional: restore local water/vegetation balance or create a safe route.
- Long-term: investigate the Sundering and access further regions (full game only).

7. Player progression, knowledge, and objectives

7.1 Philosophy

Progression should expand what the player understands and can do. Numerical upgrades may support discovery but should not replace it. Avoid a large skill tree until interaction rules are proven.

Track	Advancement	Examples	Prototype
Elemental fluency	Discover/apply reliable rules.	Improved control, safer application, new combination.	One unlock/journal milestone.
Field knowledge	Observe organisms, materials, local patterns.	Habitat clues, resource identification, recovery forecasts.	Journal + three discoveries.
Crafting capability	Find plans and process materials.	Tools, containers, shelter parts.	Small fixed recipe list.
Regional access	Complete meaningful interventions.	New site, route, specialist help.	One regional objective.
Traversal	Use elemental effects and crafted equipment.	Climb aid, glider-like device, heat protection.	One knowledge-gated route.

7.2 Field journal

The journal is both UI and gameplay system. It distinguishes observations from confirmed rules so a coincidence is not presented as proof.

Entry	Example	Behaviour
Observation	“Moss near the spring looks healthier after rainfall.”	May be incomplete or uncertain.
Hypothesis	“Moisture may improve this soil’s growth potential.”	Player-created or suggested from evidence.
Confirmed rule	“Watering compatible soil improves growth suitability.”	Added after repeated or controlled results.
Organism profile	Habitat, diet clues, territorial signs, reactions.	Fills in as the player observes.
Regional record	Changes to water, vegetation, hazards, landmarks.	Updates after meaningful state transitions.

7.3 Journal UX rules

- Record the result and likely cause, not only a recipe string.
- Filter by element, material, organism, and region.
- Do not reveal outcomes before evidence unless a chosen guidance setting requests it.
- Allow hypotheses to be corrected when later evidence disproves them.
- Use confidence labels: observed, repeated, confirmed, unknown.

7.4 Quest structure

Use problem-led objectives rather than generic errands. A regional objective asks the player to diagnose a system and choose an intervention. First major objective: restore a reliable route to a water source by understanding a local obstacle, applying at least one elemental rule, and observing a persistent change.

8. Creatures and ecosystem behaviour

8.1 Design goals

- Every creature has an ecological role: resource role, seed dispersal, grazing, predation, scavenging, or territorial pressure.
- Creatures react to a few legible inputs rather than omniscient AI.
- Environmental changes alter suitability, routes, feeding opportunities, or territory.
- Start with a small cast of distinct behaviours rather than many reskinned enemies.

Creature	Role	Behaviour	Element relationship	Priority
Mossback	Slow herbivore / seed disperser.	Grazes suitable patches, retreats from disturbance, follows food.	Water/soil affect forage; fire triggers avoidance.	High
Cinderwing	Small scavenger / airborne indicator.	Circles warm updrafts, visits disturbed ground, flees threats.	Air/heat create opportunities; smoke changes response.	Medium
Riftstalker	Territorial predator.	Patrols bounded territory, investigates stimuli, retreats if habitat is poor.	Water scarcity/habitat loss can shift patrols; fire is temporary barrier.	Later

8.2 Behaviour state machine

State	Entry	Action	Exit
Idle/Rest	No urgent need; safe.	Rest, idle animation, low-cost perception.	Need, stimulus, schedule.
Forage	Suitable food patch.	Move to patch; consume/interact.	Patch depleted, threat, need met.
Investigate	New sound, elemental event, disturbance.	Approach safe distance; inspect.	Stimulus resolved, threat, timeout.
Flee	Threat above fear threshold.	Move to safe point using navigation.	Distance threshold and threat absent.
Defend/Attack	Threat crosses territory rule.	Telegraphed warning, bounded attack/chase.	Threat retreats, stamina threshold, territory boundary.
Recover/Relocate	Habitat suitability too low.	Move to nearby suitable patch or reduce activity.	Suitable habitat or region update.

8.3 Habitat suitability

Use a weighted score from food, moisture, temperature, disturbance, and safe cover. Prototype scoring is simple and deterministic. Thresholds change behaviour; machine learning is not required.

8.4 Population persistence

Start with a fixed small population. Later, simulate population trends by habitat patch rather than keeping every individual active in unloaded regions. Regional summaries can represent few, stable, stressed, or absent without spawning dozens of actors.

8.5 Fairness

- Give visible/audio warning before a dangerous creature attacks.
- Do not let off-screen creatures alter the world in ways the player cannot discover.
- Do not make elemental rules inconsistent to protect a scripted encounter.
- Expose state, target, suitability, and intent in debug overlays.

9. Resources, crafting, and construction

Category	Examples	Acquisition	Use
Organic	Branches, fibre, resin, edible plants.	Gather, harvest, cultivate, creature interactions.	Tools, bindings, shelter, fuel.
Mineral	Stone, clay, ore, altered mineral.	Gather, expose deposits, process with tools/heat.	Foundations, tools, components.
Environmental	Clean water, moisture, heat, airflow access.	Collect, channel, store, create/protect conditions.	Survival, growth, crafting, process control.
Knowledge/plans	Rules, structure plans, material clues.	Observe, experiment, explore ruins.	Unlock capability; not physical inventory by default.

9.1 Crafting rules

- Recipes are data-driven: recipe ID, inputs, outputs, station, knowledge requirement.
- First prototype uses authored recipes; experimentation can expand later.
- Crafting should connect to elemental systems: dry fibre, alter minerals, improve soil.
- Start with approximately 6–10 useful recipes; expand only when the loop needs them.

9.2 Initial recipe candidates

Item	Inputs (provisional)	Purpose
Hand tool	Branch + stone + fibre	Basic harvesting and light obstacle interaction.
Water container	Fibre/resin or clay component	Carry water or apply controlled moisture.
Simple shelter	Branches + fibre + plant material	Rest point and partial exposure protection.
Storage rack	Branches + fibre	Organise resources.
Heat-safe vessel	Clay + fired component	Basic material processing.
Small barrier / channel marker	Stone or timber	Modify or test a flow route.
Field kit	Fibre + tool component	Support observation interactions if needed.

9.3 Building philosophy

Building is functional and modular. Early structures provide shelter, storage, work surfaces, and limited control over environmental processes. Later systems may include irrigation, wind capture, thermal processing, barriers, and habitat-support structures. Avoid complex power networks until process-building is proven fun.

9.4 Placement and validation

- Ghost preview shows footprint, orientation, collision, slope suitability, and material requirements.
- Invalid placement explains why: steep ground, blocked footprint, missing support, or missing resources.
- Use explicit tags and connection points where useful; avoid a universal freeform connection system in the prototype.
- Keep placement validation centralised so multiplayer authority can be added later if approved.

9.5 Durability

Structures should not decay merely because the player is away. Durability is reserved for meaningful hazard or interaction damage. Routine upkeep is added only if it creates a compelling planning decision.

10. User interface, feedback, and accessibility

UI element	Purpose	Priority
Element selector	Selected element, availability, cost/cooldown.	Essential
Target feedback	Whether a target is eligible and why action fails.	Essential
Health/stamina/exposure	Immediate survival state.	Essential
Interaction prompt	Context-sensitive actions without long text.	Essential
Discovery toast	Confirm new observation/rule.	Essential
Compass/landmarks	Navigation without minimap dependency.	Candidate
Journal	Searchable observations, rules, organisms, regional changes.	Essential
Build menu	Recipes, materials, placement validation.	Construction milestone

10.1 Feedback stack

Every meaningful elemental result should combine at least two signals: visible world change, audio cue, concise UI message, journal update, or clear animation. High-value discoveries deserve stronger feedback than routine updates.

10.2 UI principles

- Use icons and shapes as well as colour; never communicate element identity through colour alone.
- Keep the HUD quiet while exploring; surface detail in journal or inspection panels.
- Keep important status readable at 1080p and scalable with UI scale.
- Avoid meters for needs that do not create meaningful choices.
- Add remappable controls, subtitles, adjustable camera sensitivity, and scalable text as production matures.

10.3 Debug UI

Development builds need togglable overlays for cell variables, active rules, creature states, target eligibility, region IDs, and save versions. These tools reduce AI-assisted debugging time and distinguish gameplay bugs from visual bugs.

11. Technical implementation in Unreal Engine

11.1 Strategy

Use Unreal Engine's gameplay framework and Blueprint-first implementation for the prototype, introducing C++ only when measured needs or maintainability justify it. Prefer focused components and data assets over one giant player Blueprint.

> **Implementation principle: prove each rule in a small test map before integrating it into the valley. Keep elemental rules deterministic and keep AI tools out of the runtime dependency path.**

System	Suggested UE implementation	Responsibility
Player	Character Blueprint + focused Actor Components	Input, movement, interaction request, survival attributes.
Element system	Element component + data assets/data table	Selected element, cost, target validation, rule dispatch.
Rule resolver	Subsystem or dedicated manager + data-driven rules	Resolve compatible inputs, priority, events.
World cell state	Subsystem/manager with spatial grid or cell map	Store local variables; bounded transitions.
Target interface	Blueprint Interface + Gameplay Tags	Expose supported effects and queryable state.
Plant patches	Actor Blueprint + species data	Visual stage, growth suitability, harvest, recovery.
Creature AI	AI Controller + StateTree or Behaviour Tree + perception	Bounded state, navigation, stimulus response.
Construction	Build component + placeable actor data	Preview, validation, cost, persistence.
Journal	Data model + UMG widget	Observations, discovered rules, region records.
Persistence	SaveGame object(s) with version field	Player, structures, journal, region snapshots.

11.2 Core data structures

Data object	Minimum fields	Implementation note
ElementDefinition	ElementID, display name, icon, action profile, cost, cooldown.	Designer-editable data asset.
InteractionRule	RuleID, input elements, target tags, preconditions, outputs, priority, feedback ID.	Deterministic and debuggable.
WorldCellState	CellID, temperature, moisture, soil stability, airflow tier, vegetation/pressure summary.	Bounded values and versioned serialization.
SpeciesDefinition	SpeciesID, habitat weights, thresholds, resource outputs, visual references.	Tune without editing AI graphs.
RecipeDefinition	RecipeID, inputs, outputs, station and knowledge requirements.	Validate item IDs.
StructureDefinition	StructureID, mesh, footprint, cost, placement tags, connections.	Instance stores transform and mutable state.
JournalEntry	EntryID, type, related IDs, evidence count, confidence, region.	Persist data, not UI text alone.

11.3 Event-driven state changes

Use events such as OnElementApplied, OnCellStateChanged, OnPlantStageChanged, OnCreatureIntentChanged, and OnDiscoveryRecorded. Avoid every system polling every actor each frame. The world manager listens and schedules only necessary work.

11.4 Tick discipline

- Do not tick every plant, resource node, structure, and distant creature continuously.
- Use timers, events, distance-based activation, and batched updates.
- Separate VFX from persistent state; effects may end while world state remains.

- Use navigation/collision for local movement and placement, not as the ecosystem simulator.
- Profile packaged Development builds, not only editor sessions.

11.5 AI-assisted workflow

- Ask AI for exact asset names, variables, node flow, preconditions, failure cases, and test steps.
- Implement one small system at a time; avoid project-wide rewrites.
- Use source control before major changes and commit each working system.
- Ask for dependencies and regression tests before changing architecture.
- Keep game project files separate from employer data, credentials, and confidential information.

11.6 Source control

Create Git baseline before installing major frameworks or starter systems. Use Git LFS for large binary assets if appropriate. Exclude generated caches such as Saved, Intermediate, and DerivedDataCache. Test a clean clone before treating the repository as a reliable backup. Evaluate any starter framework in a disposable branch or copy first.

12. Data model, save system, and persistence

Data	Persist?	Method
Player location, health, unlocked elements	Yes	Player save record.
Journal discoveries and hypotheses	Yes	Entry IDs and confidence/evidence state.
Placed structures	Yes	Structure ID, transform, mutable state, contents if needed.
Changed environmental cells	Yes, if meaningful	Sparse cell records; omit default state.
Creature individual positions	Usually not initially	Recreate from region rules or save special individuals.
Particles / temporary visual effects	No	Rebuild visuals from gameplay state if needed.
Short-lived effects	Usually no	Persist only if duration changes gameplay after reload.

12.1 Sparse persistence

Store only cells/objects that differ meaningfully from authored baseline. Use stable region ID plus cell coordinate/index. Cell-grid resolution is a prototype parameter: select it by testing effect radius and required spatial precision.

12.2 Save rules

- Manual save and reliable checkpoint are sufficient for the first vertical slice.
- Every save has a schema version. Migrate known fields or reject with a clear diagnostic; never silently corrupt.
- Save after meaningful transitions and before risky experimental features; do not save the full world every frame.
- On load, restore persistent state first, then rebuild visuals and derived systems.
- Provide a development-only reset-region command for testing recovery and migration.

12.3 Multiplayer later

If multiplayer is approved, persistent environmental changes need an authoritative owner and replication strategy. Do not build full networking into the prototype, but avoid tying all state changes to local-only cosmetic actors.

13. Audio, visual direction, and presentation

13.1 Visual direction

The look is grounded, tactile, and atmospheric rather than brightly cartoonish. Use layered vegetation, worn structures, exposed rock, wet/dry material variation, atmospheric depth, and creatures whose anatomy fits their environment. Stylisation is acceptable where it improves clarity, but element effects should feel like forces acting on materials.

Element	Visual language	Audio language	UI cue
Fire	Warm light, heat distortion, embers, charred transitions.	Crackle, low roar, material response.	Flame-like icon + shape/label.
Water	Surface sheen, darkened soil, droplets, pooling.	Splash, trickle, impact variation.	Droplet/wave icon.
Earth	Dust, cracks, displaced strata, compacted surfaces.	Scrape, crumble, low thud.	Layered-ground icon.
Air	Directional foliage, drifting particles, cloth response, lift cues.	Wind tone, rustle, pressure swell.	Swirl/stream icon.

13.2 Feedback before fidelity

A readable change is more valuable than a beautiful ambiguous effect. A simple mesh state change, clear sound, and visible moisture/temperature cue are preferable to expensive particles that hide the rule.

13.3 Performance-aware art

- Use instancing for repeated foliage and props.
- Use LODs/Nanite where appropriate, but profile real content.
- Keep volumetric effects scalable; smoke, fog, and foliage can be expensive together.
- Use material parameters for wetness, burn state, and dust where practical.
- Polish one valley only after the greybox loop works.

14. Prototype scope, milestones, and acceptance tests

Do not build the whole world first. The first milestone is a small, testable elemental laboratory proving cause and effect, persistence, and player curiosity.

Milestone	Build	Acceptance evidence
M0 — Foundation	UE project, source control, input, movement, test map, naming conventions.	Clean checkout opens; player moves; a change can be reverted.
M1 — Element lab	Four selectable elements, simple target types, first deterministic interactions.	Three interactions produce visible repeatable outcomes; invalid targets explain failure.
M2 — World state	Local variables, heat/moisture/soil state, debug overlay.	Change persists for defined duration or across save/load; values remain bounded.
M3 — Ecosystem	Two plant patches and one creature with state-driven behaviour.	Water/heat/disturbance changes suitability and causes observable response.
M4 — Survival/building	Shelter, storage, health/stamina/exposure, placement validation.	Player establishes safe base and uses it to support an experiment.
M5 — Journal/objective	Observation entries, confirmed rules, one regional problem.	Tester solves objective using a learned rule and explains why.
M6 — Vertical slice	Polish, audio, onboarding, save/load, performance pass.	External testers understand premise and choose to run more experiments.

14.1 First playable content cap

Content	Maximum first-playable target
Regions	One valley; no open-world streaming requirement.
Element actions	One clear primary action per element.
Interaction rules	Six core rules; expand only if readable and fun.
Environmental variables	Temperature, moisture, soil stability, vegetation health, airflow tier.
Plant types	Two patch types with different needs.
Creatures	One or two; begin with one non-hostile ecosystem actor.
Crafting	Six to ten useful recipes.
Structures	Shelter, storage, one environmental control structure.
Regional objective	One problem with at least two plausible solutions.
Persistence	Journal, structures, meaningful local cell changes.

14.2 Playtest questions

- Can a first-time player discover a useful interaction without the exact recipe being explained?
- Can the player explain why the outcome occurred?
- Does the result create a new option, problem, or discovery?
- Does the world appear to remember what happened after leaving and returning?
- Do players naturally try multiple experiments?
- Does survival support the elemental loop or distract from it?
- Do testers describe the game as an elemental ecosystem sandbox rather than a conventional survival game?

14.3 Go/no-go

Expand only if most testers understand cause and effect, use the journal voluntarily, and try a second experiment without prompting. If not, improve feedback and reduce variables before adding creatures, regions, or recipes.

15. Risks, performance budgets, and production discipline

Risk	Impact	Mitigation
Scope explosion	Each new element, organism, and region multiplies interactions.	Cap first-playable content; expand only after tests pass.
Opaque causality	Players stop trusting the simulation.	Explicit rules, clear feedback, journal evidence, debug overlays.
Simulation cost	World, foliage, AI, VFX, and updates compete for frame time.	Aggregate distant state; timers/events; profile early.
Survival friction	Needs and gathering overshadow discovery.	Start with high-value needs; defer hunger unless justified.
AI-assisted code drift	Broad changes break systems and hide dependencies.	Small tasks, version control, acceptance tests.
Weak market identity	“Survival with elements” is generic.	Lead with ecological memory and player-created consequences.
Persistence bugs	World changes reset or corrupt saves.	Versioned sparse saves and migration tests.
AI service dependency	Cost/availability changes affect development.	Core game runs offline without runtime LLM calls.

15.1 Initial performance targets

These are provisional PC targets; validate in a packaged build on actual development hardware and representative content.

Metric	Initial target	Notes
Frame rate	Stable 60 FPS goal at 1080p.	Reduce effects/update frequency before harming rule clarity.
Frame time	16.7 ms for 60 FPS.	Measure game thread, render thread, GPU, and memory separately.
Active creatures	Start with 1–5 relevant actors.	Scale by profiling.
World simulation	Only nearby cells update frequently.	Regional summaries update in batches.
Save time	Short enough not to disrupt play; measure first.	Sparse state and bounded object counts.

15.2 Production discipline

- Maintain a working build at every milestone.
- For each system, document purpose, inputs, outputs, failure cases, test map, and acceptance criteria.
- Do not ask AI to refactor the whole project when a narrow change suffices.
- Use placeholder art until mechanics are stable; do not model a large creature roster before behaviour is proven.
- Track known issues and distinguish intended design from bugs.
- Record scope decisions in Section 16 so old ideas do not re-enter unnoticed.

15.3 Explicit non-goals for first playable

- A 1000 km² world or multiple streamed zones.
- Multiplayer, PvP, hosted servers, or live-service architecture.
- Dozens of elemental combinations.
- Procedural narrative generation or LLM-controlled creature behaviour.
- Fully deformable terrain, fluid simulation, or realistic climate modelling.
- Complete main story, extensive dialogue, or large quest chains.

- Advanced automation/power networks or hundreds of recipes.

16. Open decisions and change control

Decision	Current recommendation	Resolve when
Final title	Keep Worldweave as working title; verify availability before public use.	Before public branding/store page.
Camera	Third-person for prototype.	After movement/target readability tests.
Element delivery	Start with aimed application; abstract effect system from input.	After M1 playtests.
Costs/cooldowns	Minimal at first; tune if spam undermines decisions.	After core rules are fun.
Death penalty	Preserve journal; safe respawn.	After survival loop tests.
Multiplayer	Out of scope until single-player vertical slice validates premise.	After vertical slice/feasibility review.
World scale	One valley first; full scale later.	After simulation/streaming benchmarks.
Runtime AI	No LLM dependency for core gameplay.	Only if bounded optional value emerges.
Narrative ending	Keep cause of Sundering partially open.	During full-game narrative design.

16.1 Change control

Any change that adds a core system, expands the first-playable cap, or changes an elemental rule must state: (1) player problem solved, (2) expected benefit, (3) implementation cost, (4) systems affected, (5) acceptance test, and (6) milestone impact. A good idea is not automatically a production priority.

16.2 Definition of done

- Feature has a written purpose and bounded scope.
- Normal case and at least two failure cases are tested.
- Result is readable to the player.
- Feature survives save/load if intended to persist.
- No known severe performance/stability issue in test map.
- Project is committed and can reopen from repository.

16.3 Immediate next actions

Order	Action	Deliverable
1	Create clean UE project and Git baseline.	Versioned project that opens and runs.
2	Create greybox valley and test range.	Small map with target objects and test cases.
3	Build target interface and element selection.	Player selects/applies one element to one target.
4	Implement first three rules only.	Deterministic resolution and visible outcomes.
5	Add world-state debug overlay.	Inspect variables and rule decisions.
6	Test and document fun/confusing points.	Evidence-based decision on expanding rules.

A. Appendix A — Prototype content and interaction matrix

Target	Properties	Initial reactions
Combustible patch	Fuel tier, moisture, burn state, regrowth state.	Fire ignites if dry enough; water cools/wets; air affects spread tendency.
Soil patch	Moisture, stability, organic quality.	Water raises moisture; earth changes barrier/stability; air can dry exposed soil.
Hot surface	Temperature, heat capacity tier, material tag.	Water cools and may create steam; air alters cooling within bounds.
Loose mineral deposit	Material ID, exposed/buried state, heat response.	Earth exposes/covers; fire transforms compatible material.
Plant patch	Species ID, health stage, suitability thresholds.	Moisture/soil/temperature affect growth; fire damages; recovery over time.
Creature	Species ID, current state, habitat score, threat response.	Investigates stimuli, flees or relocates at explicit thresholds.

A.1 Initial rule matrix

Rule ID	Input	Target	Expected change	Test
ER_WATER_WET	Water	Combustible patch / soil	Moisture rises to cap; ignition eligibility may fall.	Compare dry and wet starts.
ER_FIRE_IGNITE	Fire	Dry combustible patch	Burn starts; temperature rises; fuel decreases over time.	Wet patch resists/rejects ignition.
ER_WATER_COOL	Water	Hot surface	Temperature falls; steam if threshold met.	Confirm threshold and feedback.
ER_AIR_DISPERSE	Air	Smoke/dust target	Particle volume shifts; gameplay effect bounded.	Compare calm/strong airflow.
ER_WATER_SOIL	Water + soil context	Compatible soil patch	Moisture/growth suitability rise.	Verify plant responds later.
ER_EARTH_BARRIER	Earth	Buildable flow edge	Barrier modifies simplified flow link.	No endless update loop.

A.2 Minimum test scenarios

- Water dry soil, then apply fire nearby; verify wet/dry difference is visible and deterministic.
- Water a hot surface; verify temperature change and steam feedback at the declared threshold.
- Change a cell, leave its immediate area, return, and verify state remains correct.
- Save, quit, reload; verify journal, structures, and modified cell state.
- Place a small earth barrier and verify simplified flow changes without an endless loop.
- Disable a visual effect and verify gameplay state remains correct.

B. Appendix B — Glossary and implementation vocabulary

Term	Definition
Ecological memory	Persistent environmental state that influences growth, creature behaviour, resource availability, or travel.
Elemental rule	Deterministic mapping from element/action, target context, and preconditions to a state change.
World cell	Bounded spatial unit storing aggregated gameplay variables; not necessarily a visible tile.
Suitability score	Bounded measure for whether a plant or creature can thrive in a local patch.
State transition	Change between named states when a declared condition is met.
Data-driven	Values/rules stored as editable data rather than buried in logic graphs.
Actor Component	Reusable Unreal component attached to an Actor for focused behaviour.
Blueprint Interface	Contract allowing different actors to expose the same functions without sharing a parent class.
Gameplay Tags	Hierarchical labels for targets, materials, states, and capabilities.
Subsystem	Unreal-managed object for systems needing shared access and a defined lifetime.
Sparse persistence	Saving changed/important state rather than every default world value.
Vertical slice	Small polished sample demonstrating intended experience end-to-end.
Greybox	Simple placeholder environment used to test scale, flow, and mechanics before final art.
Acceptance test	Observable condition required before a milestone is considered complete.

Closing design statement

> **Worldweave succeeds when players stop asking only “What can I craft next?” and start asking “What happens if I change this part of the world?” Every major system should reinforce that shift.**

End of GDD Version 0.1